Colina, S. 2006. "Optimality-theoretic advances in our understanding of Spanish syllabic structure"

Goal: to highlight the advantages of an OT approach to syllabification to support the claim that OT has brought about significant improvement in syllabic theory.

- replacement of language-specific rules with universal constraints;
- resolution of rule conspiracies; reglas que se aplican bajo ciertas restricciones (y otras no)
- elimination of stipulatory statements (quality of epenthetic vowels) and adhoc conditions on rule application.
- variation: various ways of obtaining the same goal (e.g., elimination of coda consonants: complete deletion, voice neutralization, stricture neutralization, vocalization, etc.). cambiando el orden de las restricciones
- separate processes in derivational models (i.e., Spanish diphthongization, resyllabification and onset strengthening) shown to respond the same motivation (avoid onsetless syllables).

0. The syllable in phonological theory...

Importance of the syllable in phonology

- allows to formulate generalizations that could not be expressed without it:
  - nasal velarization (e.g.[paŋ← /pan/ 'bread'), /s/ aspiration (e.g. [doh] ← /dos/ 'two'), /s/ voicing assimilation (e.g. [mizmo] ← /mismo/'same'), liquid gliding (e.g. /papel/ [papei̯] ← /papel/ 'paper'), nasal assimilation (e.g. [kambio] ← /kanbio/'change').
- stress generalizations (*3-syllable window*, *branching condition*, Harris 1983).
- phonotactic restrictions also hold over syllables and syllabic components (**muersto* can only be explained as the result of an excessive number of segments in the rhyme)

Syllable structure in OT

(a) *syllable structure is the result of conflicting universal requirements on syllable structure prioritized differently across languages.*

(b) syllable structure (i.e., the output of syllabification/a syllabified output) is *the result of the interaction of universal faithfulness and markedness constraints*.

(c) OT can satisfactorily explain *conspiracies*.

1. Syllable types and phonotactics in Spanish

1.1. Basic syllable types

(3) Syllable types in Spanish

V	*a*.la	'wing'
CV	*co*.lor	'color'
CVC	*pan*	'bread'
VC	*un*	'one'
CCV	*flo*.tar	'float'
CCVC	*tren*	'train'
VCC	*ins*.truir	'instruct'
CVCC	*pers*.pec.tiva	'perspective'

CCVCC *trans*.por.te 'transportation'

Universal nature of OT constraints, superior to the purely descriptive list in (3).

CV is the preferred, unmarked syllable type.

some languages allow codas, none require them

some languages allow onsetless syllables, but none forbid onsets. no languages require codas and forbid onsets.

(4) ONSET: No vowel-initial syllables
NO CODA: Syllables cannot end in a coda

Complex syllabic constituents, universally more marked than singletons:

(5) *COMPLEX ONSET: No more than one segment in the onset
* COMPLEX CODA: No more than one segment in the coda
* COMPLEX NUCLEUS: No more than one segment in the nucleus

(6) Basic segmental faithfulness constraints

MAX-IO: Every segment present in the input must have a correspondent in the output (anti deletion constraint).

DEP-IO: Every segment present in the output must have a correspondent in the input (anti epenthesis constraint).

FAITH: (cover term for all faithfulness constraints)

Basic syllable type in Spanish

FAITH >> ONSET, *CODA.

(7) /ala/ [a.la]

	FAITH	ONSET	*CODA
a. ☞ a.la		*	
b. Ta.la	*!		
c. al.a		**!	*
d. la	*!		

(8) /poner/ [poner]

	FAITH	ONSET	*CODA
a. ☞po.ner			*
b. po.ne.re	*!		
c. pon.er		*!	*
d. po.ne	*!		

(10) FAITH >> *COMPLEX ONSET, *COMPLEX CODA.

(11) /broma/ [bro.ma] ‘joke’

	Faith	*Complex Onset
a. ☞ bro.ma		*
b. ro.ma	*!	
c. be.ro.ma	*!	

(12) /insto/ [ins.to] ‘I urge’

	Faith	*Complex Coda
a. ☞ ins.to		*
b. in.to, is.to	*!	
c. is.to	*!	
d. i.nes.to	*!	
e. in.se.to	*!	

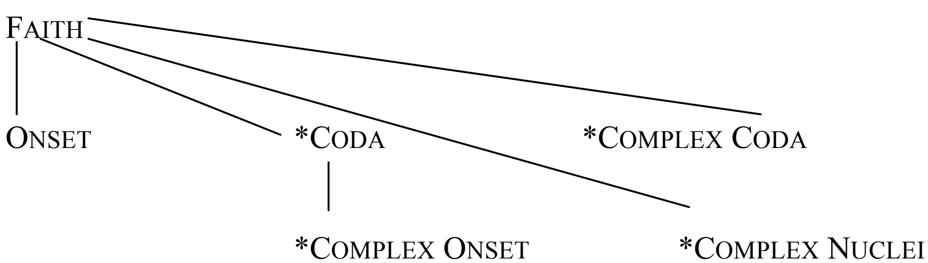
No Coda >> *Complex Onset (13) (cf. *onset maximization* in derivational approaches).

(13) /potro/ [po.tro] ‘young horse’

	*Coda	*Complex Onset
a. ☞ po.tro		*
b. pot.ro	*!	

*Complex Nuclei is violated by diphthongs, /pierde/ → [pi̯érðe].

(14) Ranking responsible for basic syllable types in Spanish



1.2 Segments and the syllable

Syllabic constraints above (referring to subsyllabic components and number of segments) are not sufficient.

1.2.1 The sonority contour

(17) Harmonic Alignment of Prominence Scales (sonority and syllabic prominence)

Nucleus >….. > Onset
| |
Vowel >….. > Obstruent

(18) Constraint Hierarchy
*NUC/obstruent >> …….>> *NUC/vowel
*ONSET/vowel >> …….>> *ONSET/obstruent

(19) Segment types in the nucleus
*NUC/obstruent >> *NUC/nasal >> *NUC/liquid >> FAITH, *NUC/vowel

(21) Segment types in the onset
*ONSET/vowel >> FAITH >> *ONSET/ liquid >> *ONSET/ nasal >> *ONSET/obstruent

(22) Segment types in the coda
*CODA/vowel>> FAITH>> *CODA/obstruent>> *CODA/nasal>> *CODA/liquid>> *CODA /glide

Pre-vocalic glides

(23) *ONSET/glide, ONSET >> *COMPLEX NUCLEUS, *NUC/glide

(24) /pierde/[pi̯érðe] 'he looses'

	*ONSET/glide	ONSET	*COMPLEX NUCLEUS	*NUC/glide
a. ☞ pi̯ér.ðe			*	*
b. pi.ér. ðe		*!		
c. pjér. ðe	*!			
d. pj.ér. ðe		*!		*

Onset glides

(26) IDENT (son): The specification for the feature [sonorant] present in the input must match the one in the output.

(27) Onset strengthening
*ONSET/glide, ONSET >> MAX-IOμ, IDENT (son)

(28) /kre-iendo/ [kre.yen̪.do]

	*ONSET/glide	ONSET	MAX-IOμ	IDENT (son)
a. ☞kre.yen̪.do			*	*
b. kre.i̯én̪.do		*!	*	
c. kre.jén̪.do	*!		*	
d. kre.i.én̪.do		**!		

Solves problems of causality of serial models.

Sonority of non-nuclear clusters: sonority distance constraint (Maximum Sonority Distance [MSD]). he maximum sonority distance between possible onset segments.

1.2.2 Segments and syllabic positions

1.2.2.1 Segments in the syllable coda
1.2.2.1.1 Word-medial codas
Derivational models: Coda Rule: "adjoin a coda to the right of the nucleus. *Possible codas are….* ". OT does not need to do this.

Spanish has codas, *CODA must be dominated. Obstruents make the worst codas.
(31) *CODA /obstruent >> *CODA /nasal >> *CODA /liquid >> * CODA /glide

Dialects with obstruent deletion
(32) *CODA /obstruent, DEP-IO >> MAX-IO

(33) MAX-IO: Every segment present in the input must have a correspondent in the output
DEP-IO: Every segment present in the output must have a correspondent in the input

(34) /obsoleto/ [osoleto]

	*CODA /obstruent	DEP-IO	MAX-IO
a. ☞ o.so.le.to			*
b. ob.so.le.to	*!		
c.o.be.so.le.to		*!	

Formal varieties without deletion
(36) /obsoleto/ [obsoleto]

	MAX-IO	DEP-IO	*CODA/obstruent
a. o.so.le.to	*!		
b. ☞ ob.so.le.to			*
c.o.be.so.le.to		*!	

Featural neutralizations in voice and continuancy

(37) IDENT(voice): A segment's input specification for [voice] must match that of the output.
IDENT(continuant): A segment's input specification for [continuant] must match that of the output.
(38) Obstruent devoicing in coda position
*SONORANT [-voice] >> *CODA [+voice] >> IDENT (voice)

(39) Obstruent devoicing in coda position
/digno/ [dikno]

	* CODA [+voice]	IDENT (voice)
a. ☞ dik.no		*
b. dig.no	*!	

Voice assimilation

(40) AGREE (voice):adjacent segments share the same specification for the feature (voice) (Lombardi 1999)

AGREE (voice) >> IDENT (voice)

(41) Regressive voice assimilation of coda obstruents

/futbol/ [fuðβol]

	AGREE (voice)	IDENT(voice)
a. ☞ fuðβol		*
b. futβol	*!	

(42) Different outcomes of voice neutralization in coda obstruents: Coda devoicing

/futbol/ [futβol]

	*CODA [+voice]	AGREE (voice)	IDENT (voice)]
a. fuðβol	*!		*
b. ☞ futβol		*	

(43) Different outcomes of voice neutralization in coda obstruents: Voice assimilation

/futbol/ [fuðβol]

	AGREE (voice)	* CODA [+voice]	IDENT(voice)
a. ☞ fuðβol		*	*
b. futβol	*!		

No conflict

(45) /obsoleto/ [opsoleto]

	AGREE (voice)	*CODA [+voice]	IDENT(voice)
a. ☞ opsoleto			*
b. obsoleto, oβsoleto	*!	*	

(46) /obsoleto/ [opsoleto]

	*CODA [+voice]	AGREE (voice)	IDENT(voice)
a. ☞ opsoleto			*
b. obsoleto, oβsoleto	*!	*	

Neutralization in continuancy, surfacing as either stops or fricatives.

(47) Markedness >> IDENT(continuant)

*CODA/ stop>> *CODA/ fricative, IDENT(continuant)

Preservation of place features in obstruents: /fuDbol/~ [fuðβol] ~ [futβol] *[fupβol] *[fuββol]; /oBsoleto/ [opsoleto] ~ [oβsoleto] *[oðsoleto] ~ *[otsoleto].

(49) IDENT^{OBSTR}(place): The place features of an obstruent in the input must match that of the output.

CODA COND: A coda cannot license place features

IDENT^{OBSTR}(place)>> CODA COND

(50) /obsoleto/ [opsoleto]

	IDENT^{OBSTR}(place)	CODA COND	IDENT(place)
a. ☞opsoleto		*	
b. otsoleto	*!	*	*

(51) HAVE PLACE: all segments must have place features

HAVE PLACE >> CODA COND

(52) a. Place Hierarchy: *DOR >> *LABIAL >> *CORONAL

b. HAVE PLACE >> *DOR >> *LABIAL >> *CORONAL >> IDENT(place)

Nasals and laterals take their place features from the following C through assimilation, thus satisfying CODA COND and HAVE PLACE simultaneously (place features are obtained and licensed through the onset).

(53) /tango/ [taŋgo]

	IDENT^{OBSTR}(place)	HAVE PLACE	CODA COND	IDENT(place)
a. ☞ taŋgo				*
b. tango			*!	
c.tamgo			*!	*

1.2.2.1.2 Word-final codas

In word-final position only coronal consonants are possible: the sonorants /-l/, /n/, and /-r/ (*papel*, *camión*, *amor*) and the obstruents /-s/ and /θ/ (*mes*, *pez*), and /d/ (virtud). /d/ can be realized as [ð] , [θ], [t] or be deleted. Other consonants appear in unassimilated borrowings, *club, frac, bulldog, album, chef* .

(54) Non-coronal coda obstruents are deleted (rather than altering place features). Coronal coda obstruents are retained

*CODA obstruent, DEP-IO >> MAX-IO

IDENT^{OBSTR}(place), HAVE PLACE >> *DOR >> *LABIAL >> MAX-IO>> *CORONAL, IDENT(place)

(55) Preservation of the place features of word-internal non-coronal coda obstruents in latinate words (formal context)

IDENT^{OBSTR}(place), HAVE PLACE, MAX-IO, DEP-IO >> *DOR >> *LABIAL >> *CORONAL, IDENT(place) (latinate words)

1.2.2.1.3 Complex codas

glide + /s/, *dais* 'you give', or a glide + coronal sonorant, *veinte* 'twenty', *aunque* 'although'
C + /s/ *biceps* 'biceps', *torax* 'thorax', *transporte* 'transport' fast speech: [bises], [toras], [trasporte], suggesting the ranking:

(56) *COMPLEX CODA >> MAX-IO

selection of target for deletion:

(57) *COMPLEX CODA >> MAX-IO, *CODA/obstruent >> * CODA /nasal >> * CODA /liquid >> *CODA/glide

2. Syllabification rules. Resyllabification.

Figure 1

Spanish syllabification rules (Hualde 1991)

1. Node projection: Mark vowels as syllable-heads, create N nodes, and project N' and N" nodes.
2. Complex Nucleus: Adjoin a prevocalic glide under the N node.
3. CV rule: Adjoin a consonant to the left of the nucleus under the N" node.
4. Complex onset: Adjoin a second consonant under the N" node if the result would be a permissible onset cluster (stop or /f/ + liquid, except */dl/ and (*) /tl/).
5. Coda rule: Adjoin a segment to the right of the nucleus under N'.
6. Complex coda: Adjoin a consonant to the right of a glide under N'

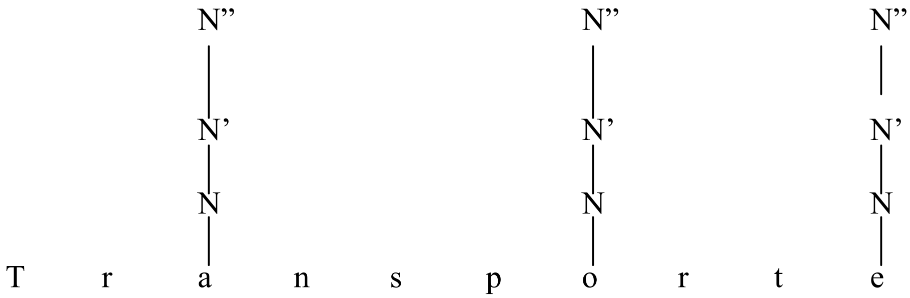
/s/-Adjunction: Adjoin /s/ under N'.

The CV rule has a second, structure-changing application. As mentioned above, an additional rule simplifying CsC clusters is also necessary (deletes a stop in a stop + /s/ cluster).

Figure 2

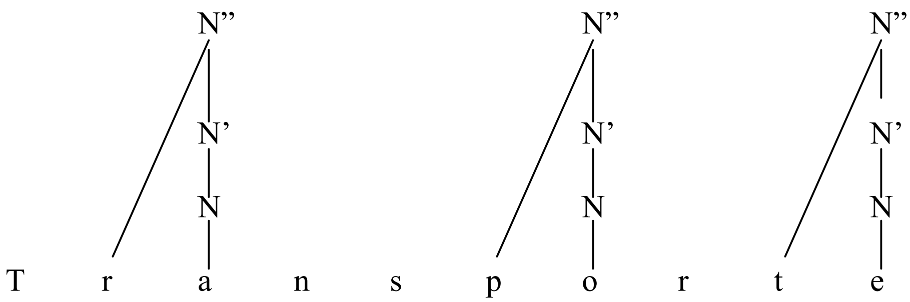
Application of Spanish syllabification rules *(Hualde 1991)*

Rule 1

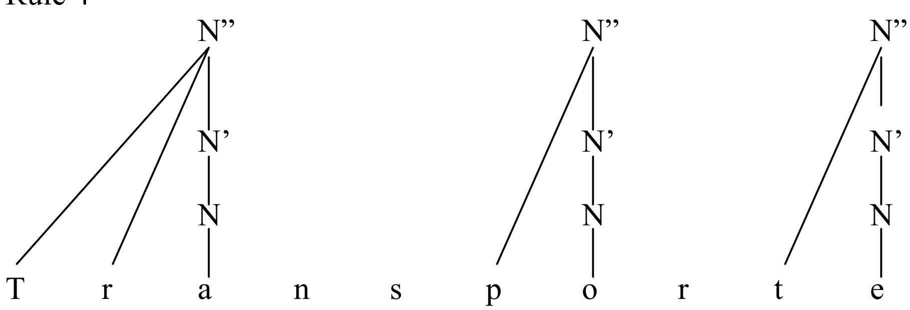


Rule 2 (N/A)

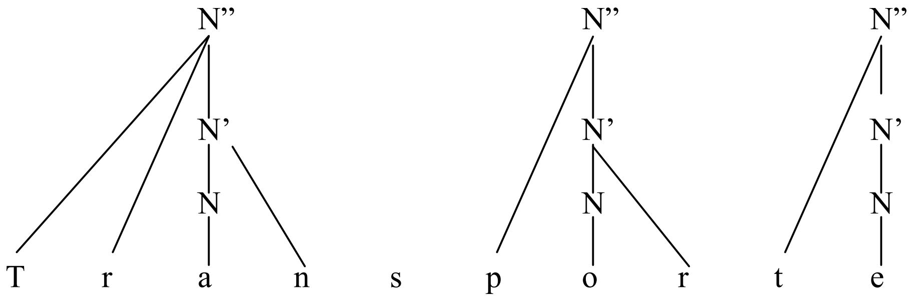
Rule 3



Rule 4

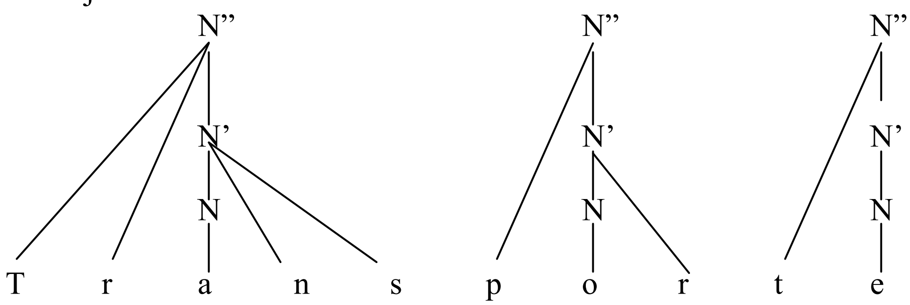


Rule 5



Rule 6 (N/A)

/s/-Adjunction



Issues:

- language-specific rules
- language-specific, extrinsic rule-ordering
- cross-linguistically common generalizations (Onset Rule), only coincidental
- syllabification algorithm requires descriptive generalizations/conditions on rule application

<u>Resyllabification</u>

(58) los amigos [lo.sa.mi. os] 'the friends'
más osos [má.so.sos] 'more bears'
reloj azul [re.lo.xa.sul] 'blue watch'

Harris (1983)

(59) Resyllabification: [+cons] --→ [+cons] / ____#V
| |
coda onset

Hualde (1991)

(60) Syllabification (including CV rule) [mas] [o.sos]

Postlexical component [mas.o.sos]
Postlexical application of CV rule [ma.so.sos]

No onset maximization (clusters) across words

(61) Hablamos [a.βla.mos] *[aβ.la.mos] 'we talk'

(62) Club lindo [kluβ.lin.do] *[klu. βlin.do] 'pretty club'

Hualde (1991) : Onset rule applies lexically and postlexically, the Complex Onset rule only has a lexical application. Why?

OT: while resyllabification of a single consonant across words is necessary to satisfy ONSET, this is not the case with the second consonant of a cluster.

(63) ALIGN (Stem, L, Syllable, L) (ALIGN-LST): Align the left edge of the syllable with the left edge of the stem

(64) ONSET >> ALIGN (Stem, L, Syllable, L)

(65) /masosos/ [ma.so.sos]

	ONSET	ALIGN-LST
a. ☞ ma.s |o.sos		*
b. mas. |o.sos	*!	

(66) ONSET >> ALIGN (Stem, L, Syllable, L) >> NO CODA

(67) /klublindo/ [kluβ.lin.do]

	ONSET	ALIGN-LST	*CODA
a. ☞kluβ. | lin.do			*
b. klu. β | lin.do		*!	

Diphthogization also responds to ONSET over ALIGN-LST and MAX-IO. Common goal

(68) ONSET >> ALIGN-LST, MAX-IOμ.

(69) /miamigo/ [mi̯a.mi.ɣo]

	ONSET	ALIGN-LST	MAX-IOμ.
a. ☞ mi̯a.mi.ɣo		*	*
b. mi.| a.mi.ɣo	*!		*

3. Domain of syllabification. Interaction with morphology

Hualde (1989b, 1991): domain of syllabification in Spanish is a unit smaller than the word affixation).

OT: domain obtained directly from the general mechanisms.

(70) / des+ ielo / [desyelo]

	ALIGN-LST
a. ☞ des. | yelo	
b. de. s|i̯e.lo	*!

los constituyentes tienen que alinearse
En este caso, el extreme izquierdo de la raíz tiene que estar alineado con el extremo derecho de la sílaba

(71) / des+ ielo / [desyelo]

	*ONSET/glide	ONSET	ALIGN-LST	MAX-IOμ	IDENT (son)
a. ☞ des. | yelo				*	*
b. de. s|i̯e.lo			*!	*	
c. des. | jelo	*!			*	*

ALIGN-LST also selects the right candidate in forms like /sublunar/ [suβlunar].

(72) /sublunar/ [suβ.lunar]

	ONSET	ALIGN-LST	*CODA
a. ☞ suβ. | lu.nar			*
b. su. β | lu.nar		*!	

EN vez de decir que lo resilabificamos en un momento y luego volvemos para agregar el prefijo… en OT alineamos (right align, left align)

That syllabification applies after suffixation shows the effects of ALIGN -R (73)

(73) ALIGN (Word, R, Syllable, R) (ALIGN -R): The right edge of a morphological word coincides with the right edge of a syllable.

4. Syllable-repair mechanisms

Syllable repair: a consequence of the basic mechanism that drives the phonology-- producing the best output possible.

OT can account for the quality of the epenthetic vowel, rather than stipulating it for each language (Spanish = [e]).

(74)

	i	u	e	o	a
high	+	+	-	-	-
low	-	-	-	-	+
back	-	+	-	+	-
round	-	+	-	+	-

Ya no se repara nada porque se hace en paralelo. La idea de la reparación es del modelo anterior.

marked, contrastive vowel features: [+low], [+round], [+high]. [+back]

epenthetic segments are by definition absent from the input; faithfulness constraints are trivially satisfied; markedness constraints become relevant. epenthetic vowel should consist of the least marked vocalic features.

(75) IDENT (low), IDENT (round), IDENT (high), IDENT (back) >> *[+low], *[+round], *[+high] *[+back]

output must satisfy *[+low], *[+round], *[+high] *[+back].
Hence[-low], [-high], [-back], [-round] that is, [e], the default epenthetic vowel in Spanish.